

RED DOG DOCK, AK, USA

WMO: 997704

Lat: 67.575N

Lon: 164.067W

Elev: 3

StdP: 101.29

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.4	(c) -30.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 22.6	(k) -7.8	(l) 19.8	(m) -5.6	(n) 2.8	(o) 140	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.4	(c) 19.2	(d) N/A	(e) 17.3	(f) N/A	(g) 15.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.6	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.0	(b) 15.9	(c) 13.6	(e) -35.2	(f) 24.2	(g) 3.5	(h) 3.5	(i) -37.7	(j) 26.7	(k) -39.7	(l) 28.8	(m) -41.7	(n) 30.8	(o) -44.2	(p) 33.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.4	-15.9	-14.0	-14.5	-8.0	1.2	6.3	11.1	10.4	6.1	-1.5	-9.7	-13.3
	DBStd	11.60	9.71	8.91	7.19	6.40	4.98	4.04	3.44	3.28	4.12	3.91	6.00	7.50
	HDD10.0	5037	803	671	758	541	276	126	27	34	128	357	592	723
	HDD18.3	7947	1062	904	1017	791	533	360	227	247	367	615	842	982
	CDD10.0	134	0	0	0	0	2	16	59	46	11	0	0	0
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	7	0	0	0	0	0	1	6	1	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	4.9	5.5	6.9	5.3	4.2	4.3	3.2	4.1	4.1	4.8	5.3	5.3	6.4
Precipitation	PrecAvg	452	32	35	26	21	17	20	49	72	50	60	39	31
	PrecMax	630	68	111	126	55	46	37	98	155	111	116	116	78
	PrecMin	324	7	5	6	3	1	9	12	14	13	24	6	10
	PrecStd	79	16	26	24	15	12	8	20	33	21	24	25	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	3.4	1.8	6.8	15.6	20.4	23.7	21.3	16.8	8.3	4.0	3.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	1.8	1.4	0.3	4.6	12.6	16.9	19.7	18.5	14.4	6.2	2.3	1.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	-0.4	-0.1	-1.6	3.1	10.3	14.6	17.7	16.6	12.9	5.1	0.7	-1.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	-2.6	-1.5	-3.4	1.5	7.9	12.3	15.9	15.2	11.7	3.9	-1.1	-3.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.1	7.7	8.4	7.0	6.6	5.4	5.2	5.9	4.9	5.2	6.1
		MCDBR	6.8	6.1	6.9	8.4	10.3	9.9	8.7	7.2	5.9	4.2	4.1	5.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.173	0.237	0.259	0.326	0.365	0.345	0.356	0.362	0.311	0.252	0.171	N/A
	taud		1.898	2.082	2.217	2.057	2.085	2.399	2.444	2.422	2.505	2.304	1.952	N/A
	Ebn at Noon		154	581	797	811	816	858	835	784	748	578	146	0
	Edh at Noon		10	49	76	120	132	100	93	85	61	39	9	0
All-Sky Solar Radiation	RadAvg		0.05	0.57	2.33	4.54	5.72	6.14	4.79	3.50	2.12	0.73	0.11	0.00
	RadStd		0.01	0.07	0.18	0.24	0.54	0.53	0.50	0.44	0.23	0.09	0.01	0.00

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+1.43	N/A	N/A	+1.00	+0.82	N/A	-475	-510	+65	+1

Nomenclature: See separate page